7 STRINGS

A string is a data type that is used to represent a sequence of characters, such as numbers, text, spaces and punctuation. Strings must be enclosed in quotation marks to distinguish them from variable names. A string could be a user name, a whole sentence or paragraph of data.

LEARNING OBJECTIVES

- Describe what a string is and explain what strings are used for
- Use iteration to traverse a string
- Concatenate and split strings

SUBJECT VOCABULARY

string a sequence of characters. They can be letters, numbers, symbols, punctuation marks or spaces

A character is one of the four basic data types. It can be a single letter, a symbol, a number or even a space. A sequence of characters is called a **string**. Although strings can contain different sorts of characters, including numbers, they are all treated as if they were text.

When a computer executes a program, it needs a way of telling the difference between a string and an instruction. In most programming languages, this is achieved by enclosing strings in quotation marks (e.g. 'johnsmith@mail.com', '10/04/15' or '123'). Both single ' ' and double " " quotation marks are acceptable, as long as they are used in the same way each time. In the example, "It's her book", the apostrophe is treated as part of the string because double quotes have been used to enclose the string.

Strings are very useful when communicating with users. For example, asking them to enter some information into a program or displaying the output of a program in a format that humans can read and understand.

STRING INDEXING

GENERAL VOCABULARY

reference to refer to something

Each character in a string has an index number, with the first character at position 0. You can use the index to **reference** individual characters in a string.

Therefore the index position of the letter 'm' is 2, even though it is the third character, as shown in Table 2.6.

INDEX	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
String	C	o	m	p	u	t	e	r		S	c	i	e	n	c	e

▲ **Table 2.6** An example string index

KEY POINT

Computers start counting at 0. Therefore, although the length of the string 'Computer Science' is 16, the indexes of the characters range from 0 to 15.

LENGTH

SUBJECT VOCABULARY

function a subprogram that performs a specific task and returns a value to the main program. High-level programming languages have a number of useful built-in functions. You can also create your own or use functions available in online libraries

The Pearson Edexcel pseudocode has a built-in `LENGTH` **function**, which you can use to find the number of characters in a string. Therefore:

```
SET numChars TO LENGTH(myText)
SEND numbChars TO DISPLAY
```

would print '16'. Table 2.7 gives the method for Python, Java and C#.

LANGUAGE	METHOD	EXAMPLE	OUTPUT

Python	The function to find the length of a string is `len()`.	`string = 'Computer Science'`
		`length = len(string)`
		`print(length)`
			16

Java	Each string object has a `.length()` method.	`String s = "Computer Science";`
		`int length = s.length();`
		`System.out.println(length);`
			16

C#	The length of a string can be found by accessing the length property.	`string subject = "Computer Science";` `Console.WriteLine(subject.Length);`	16

▲ **Table 2.7** Finding the length of a string in Python, Java and C#.

SKILLS: PROBLEM SOLVING, CREATIVITY, DECISION MAKING

ACTIVITY 12

CHECKING PASSWORD LENGTH

Create and write a program to check the length of a password. If the password entered is less than six characters, the program should output 'The password you have entered is not long enough'; otherwise it should output 'Length of password OK'.

STRING TRAVERSAL

SUBJECT VOCABULARY

string traversal using a loop to cycle through each character in a string

You can use a `FOR` loop to cycle through each of the characters in a string. This is known as **string traversal**.

The following algorithm prints out the word 'monkey' letter by letter, displaying each letter on a separate line.

```
SET animalName TO 'monkey'
FOR index = 0 TO LENGTH(animalName) - 1
    SEND animalName[index] TO DISPLAY
END FOR
```

The loop runs until `LENGTH(animalName) - 1` because string indexing starts at 0 and therefore has to print characters with indexes of 0 to 5.

PYTHON

```
animalName = 'monkey'
for index in range 0, len(animalName):
    print(animalName[index])
```

Using Python you should not use `len(animalName) - 1` as the range command does not include the end number of the loop.

JAVA

```
class Main {
  public static void main(String[] args) {
    String animalName = "monkey";
    for(int index = 0; index < animalName.length(); index++) {
      System.out.println(animalName.charAt(index));
    }
  }
}
```

C#

```
string animalName = "monkey";

for (int index = 0; index < animalName.Length; index ++)
{
Console.WriteLine(animalName[index]);
}
```

SKILLS PROBLEM SOLVING, CREATIVITY, DECISION MAKING

ACTIVITY 13

WHICH CAR?

Write a program that will check if a make of car entered by the user is in the string 'The cars present included Ford, Mercedes, Toyota, BMW, Audi and Renault.'

If the car entered by the user is present, then 'It is present' should be returned or 'It is not present', if not.

It should not matter which case the car name is entered by the user.

OTHER WAYS TO MANIPULATE STRINGS

FINDING A CHARACTER WITH A PARTICULAR INDEX

The index of a character is the position of that character within a string. Note that indexing starts at 0 not 1. This means the first character will have an index of 0 and the second character will have an index of 1.

PYTHON

Python uses square brackets to access elements in a string:

```
string = 'Computer Science'
print(string[11])
```

This code would return the character 'c'.

JAVA

Java allows you to get a character at a particular index using the `charAt` method.

```
class Main {
  public static void main(String[] args) {
    String string = "Computer Science";
    System.out.println(string.charAt(11));
  }
}
```

This code would display the character 'i' (the 12th character at index 11).

C#

```
string text;
text = "Computer Science";
Console.WriteLine(text[3]);
```

This code would return the character 'p'.

CHANGING ALL CHARACTERS TO LOWER CASE

PYTHON

```
string = 'Computer Science'
string = string.lower()
print(string)
```

This code would return 'computer science'.

JAVA

```
class Main {
  public static void main(String[] args) {
    String string = "Computer Science";
    string = string.toLowerCase();
    System.out.println(string);
  }
}
```

C#

```
string text;
text = "Computer Science";
text = text.ToLower();
Console.WriteLine(text);
```

Would output 'computer science'.

CHANGING ALL CHARACTERS TO UPPER CASE

PYTHON

```
string = 'Computer Science'
string = string.upper()
print(string)
```

This would return 'COMPUTER SCIENCE'.

JAVA

```
class Main {
  public static void main(String[] args) {
    String string = "Computer Science";
    string = string.toUpperCase();
    System.out.println(string);
  }
}
```

C#

```
string text;
text = "Computer Science";
text = text.ToUpper();
Console.WriteLine(text);
```

This would output 'COMPUTER SCIENCE'.

HINT

Changing strings to upper or lower case is useful when you are comparing two strings and the case does not matter, only the characters.

EXTRACTING CHARACTERS FROM A STRING

PYTHON

```
string = 'Computer Science'
substring = string[3:6]
print(substring)
```

This code would return 'put' as it selects the characters from index 3 to index 5. Index 6 is not included.

JAVA

```
class Main {
  public static void main(String[] args) {
    String string = "Computer Science";
    String substring = string.substring(3,6);
    System.out.println(substring);
  }
}
```

C#

```
string text;
text = "Computer Science";
text = text.Substring(3,4);
Console.WriteLine(text);
```

This would display 'pute' as it selects 4 characters starting at index 3 (remember that the first character is at index 0).

CHECKING A PHRASE IN THE STRING

These examples check to see if the string 'Computer Science' contains the strings 'put' and 'PUT': 'put' is a substring of 'Computer Science' but 'PUT' is not, so these examples will display 'True' then 'False'.

PYTHON

```
string = 'Computer Science'
present = 'put' in string
print(present)
present = 'PUT' in string
print(present)
```

JAVA

```
class Main {
  public static void main(String[] args) {
    String string = "Computer Science";
    boolean present = string.contains("put");
    System.out.println(present);
    present = string.contains("PUT");
    System.out.println(present);
  }
}
```

C#

You can check if a substring is present in a string using the contains method. If the substring is present in the string then 'True' is returned, otherwise 'False' is returned.

```
string text;
bool found = false;
text = "Computer Science";
found = text.Contains("put");
Console.WriteLine(found);
```

This would display 'True', as 'put' is in 'Computer Science'.

CONCATENATION

Concatenation involves joining two or more items of information together. Concatenating two strings produces a new string object. It is very useful when displaying text on screen.

In Pearson Edexcel pseudocode concatenation is done in the following way.

```
RECEIVE userName FROM (STRING) KEYBOARD
SEND 'Hello' & userName TO DISPLAY
```

Note that literal text is enclosed in speech marks but the variable name is not. In Python, this would be:

```
userName = input('Please enter your username')
print('Hello' + userName)
```

The '+' character is used for concatenation.

In Python, you cannot concatenate strings with numbers. For example, the following would produce an error message.

```
length = 13
print('The length is' + length)
```

To overcome this, the number must be converted to a string:

```
length = 13
print('The length is' + str(length))
```

In Java, you can concatenate strings with most other data types:

```
Scanner scanner = new Scanner(System.in);
System.out.println("Please enter your username:");
String username = scanner.nextLine();
System.out.println("Hello" + username);

int length = 13;
System.out.print("The length is" + length);
```

In C# this would be:

```
string userName;
Console.WriteLine("Please enter your username:");
userName = Console.ReadLine();
Console.WriteLine("Hello" + userName);
```

In C#, you can concatenate with numbers.

COMPARING STRINGS

Comparing strings in Python and C# can be done using the == operator but in Java you need to use a string object's equals method:

```
String food = "pie";
if(food.equals("pie")) {
      System.out.println("Yum! I like pie");
}
```

SKILLS PROBLEM SOLVING, CREATIVITY, DECISION MAKING

ACTIVITY 14

CONCATENATING AND SLICING STRINGS

A company wants a program to generate usernames for new employees. Each username consists of the first four letters of the employee's last name and the first letter of their first name joined together. If the employee's last name is less than four characters in length a letter 'X' is used to fill in for each of the missing characters. Develop a program that asks the user to input their first and last names and outputs their username.

CHECKPOINT

Strengthen

SKILLS CRITICAL THINKING

S1 How are individual characters in a string referenced?

SKILLS PROBLEM SOLVING, CREATIVITY

S2 Develop an algorithm that uses a loop to traverse a string.

SKILLS CRITICAL THINKING

S3 How are a string and a non-string concatenated?

Challenge

SKILLS PROBLEM SOLVING, CREATIVITY

C1 Develop a program that asks the user to input a sentence and then splits it up wherever a space occurs. Each word should then be displayed on a separate line.

How confident do you feel about your answers to these questions? If you're not sure you answered them well, have another go at the activities in this section.

SUMMARY

- A string is a sequence of characters.
- Each character in a string has a unique index value representing its position in the string. The first character in a string has the index value 0.
- High-level programming languages have a built-in length function that finds the length of a string.
- A loop is used to traverse a string, character by character.
- Concatenation is the process of joining two or more strings together.
- Slicing is the process of extracting part of a string.
- String formatting is used to control the way text is displayed on screen.